

Reproducing the NMM2 PING bifurcation (Eq. 7.9 / Fig. 7.3)

for the TI envelope-demodulation note TN0484

To: Raul de Palma **From:** Giulio Ruffini **Re:** Rosetta Stone NMM2-PING
Ask: the exact synaptic kernel / gains, and the source code (AUTO-07P and/or a time integrator) used for Fig. 7.3.

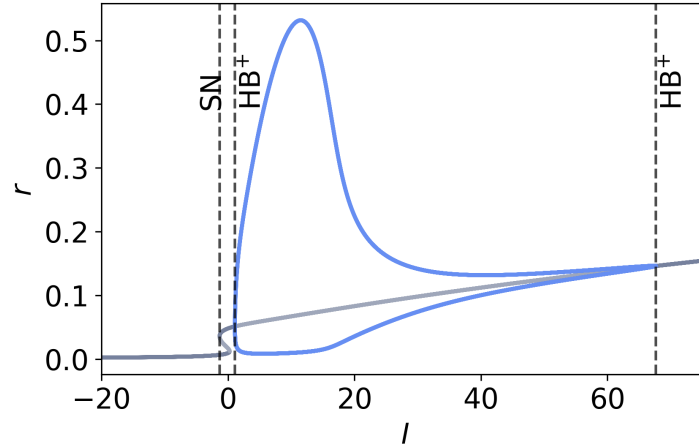
1 Context

For TN0484 we want the **NMM2 PING** as a third model—after a single Jansen–Rit column and the LaNMM—demonstrating the same mechanism: a weak amplitude-modulated (AM) electric field $I(t) = p + \lambda E(t)$ driven into the excitatory current is demodulated by the exact quadratic v^2 term and resonantly amplified near a Hopf bifurcation (read out as a lock-in at the envelope frequency over a $(\Delta f, I)$ plane). For the resonance claims to be quantitatively honest we need the PING model to be *faithful* (correct Hopf locations and gamma frequency). We reconstructed it from the Rosetta Stone but our reconstruction does **not** reproduce your published bifurcation diagram (Fig. 7.3), so we would rather start from your exact equations/code.

2 Target: the published PING bifurcation (Fig. 7.3, Eq. 7.9)

The diagram shows, as the excitatory input I increases: a saddle-node near $I \approx -1$; a lower supercritical Hopf $\text{HB}^+ \approx 1$; a **large gamma limit cycle** peaking at $r \approx 0.53$ near $I \approx 11$; an upper $\text{HB}^+ \approx 68$; and a stable focus outside. Parameters (Table A.1, NMM2-PING), with $I \in [-20, 75]$:

$$\bar{\eta}_e = \bar{\eta}_i = 0, \quad J_{ee} = J_{ei} = J_{ie} = 1, J_{ii} = 2, \quad \tau_e = 15, \tau_a = 10, \tau_i = 7.5, \tau_g = 2.5, \quad \Delta_e = \Delta_i = 1.$$



Rosetta Stone, Fig. 7.3 (reproduced): NMM2-PING bifurcation in the excitatory input I . Grey = fixed points (dark stable, light unstable); blue = limit-cycle max/min. Saddle-node near $I \approx -1$, lower $\text{HB}^+ \approx 1$, gamma limit cycle peaking $r \approx 0.53$ near $I \approx 11$, upper $\text{HB}^+ \approx 68$, stable focus outside. *This is the target our reconstruction fails to reproduce.*

3 Our (corrected) reading of Eq. 7.9

We now use the equations exactly as written in Eq. 7.9 — in particular the (r, v) membrane equations are **bare** (no τ on the left-hand side), and the cross-couplings are the off-diagonal J 's,

$C_{xy} = J_{ei} (I \rightarrow E)$, $C_{yx} = J_{ie} (E \rightarrow I)$, with $J_x = J_{ee}$, $J_y = J_{ii}$:

$$\begin{aligned} \dot{r}_x &= \Delta_x/\pi + 2r_x v_x, & \dot{v}_x &= v_x^2 + \bar{\eta}_x + J_x s_x - \pi^2 r_x^2 - C_{xy} s_y + I + \hat{F}_{e;x}, & s_x &= \hat{K}_x[r_x], \\ \dot{r}_y &= \Delta_y/\pi + 2r_y v_y, & \dot{v}_y &= v_y^2 + \bar{\eta}_y - J_y s_y - \pi^2 r_y^2 + C_{yx} s_x, & s_y &= \hat{K}_y[r_y]. \end{aligned}$$

Corrections we made after re-reading Eq. 7.9: we had wrongly divided the membrane equations by a membrane time τ_e/τ_i (there is none in Eq. 7.9), and we confirmed the $C = J_{\text{off-diagonal}}$ mapping. With these fixes all four time constants must live *only* inside the synaptic kernels $\hat{K}_x, \hat{K}_y$.

4 The remaining mismatch

Even with the bare-membrane form above, a long-settle time-integration:

- shows **no clean limit cycle** anywhere in $I \in [-20, 75]$;
- has fixed-point firing rates that climb to $r \approx 1-2$ (your branch stays $\approx 0.1-0.15$): the scale is $\sim 10\times$ too high;
- **diverges** (overflow) at larger I .

We suspect the missing pieces are (i) the synaptic kernel definition (a *gain* we are omitting and/or the rise/decay assignment of the two time constants) and (ii) the integration method—Fig. 7.3 looks like an AUTO-07P continuation, whereas our explicit RK4 blows up on the stiff MPR spikes.

5 Questions

Q1. Synaptic kernel $\hat{K}_x, \hat{K}_y$. The review defines $\hat{K}$ via $\gamma^{-1}(\tau^2 \ddot{s} + 2\tau \dot{s} + s) = r$ (equal rise/decay), or (τ_r, τ_d) for distinct rise/decay. For NMM2-PING:

- what is the synaptic **gain** γ for each population? (Table A.1 lists the J 's, τ 's and Δ but *no* γ — is $\gamma = 1$, a JR-style value, or folded into J ?)
- do the two constants map as $\hat{K}_x = (\tau_e, \tau_a)$ and $\hat{K}_y = (\tau_i, \tau_g)$, and which is rise vs. decay? Is the kernel the bi-exponential $(\tau_d \frac{d}{dt} + 1)(\tau_r \frac{d}{dt} + 1)s = \gamma r$, or critically damped with a single τ (then what are the other two constants)?
- is the DC gain of $\hat{K}$ unity ($s \rightarrow r$ at steady state), or γ ?

Q2. Units / scale. What are the units of r (Fig. 7.3 axis 0–0.5: kHz? normalized?) and the time unit, so the gamma frequency lands correctly (~ 40 Hz)? Our r runs $\sim 10\times$ high, pointing to a missing gain/normalization.

Q3. Source code (the key ask). Could we get whatever produced Fig. 7.3 — the **AUTO-07P** files + constants (the table cites <https://github.com/pclus/auto-tutorial>) and/or a robust **Python time-domain integrator** for Eq. 7.9? With that we can reproduce Fig. 7.3 exactly and then add the weak-field AM-drive resonance analysis for TN0484.

Once we have a faithful PING we will drive $\hat{F}_{e;x}$ with $A[1 + \cos \Omega t] \cos(\omega_c t)$ and read the lock-in of r_x at Ω over $(\Delta f, I)$, showing forced resonance below the Hopf and entrainment within the cycle — the gamma analogue of the JR/LaNMM alpha results already in TN0484.